

Multilayer Perceptron Training Process

COMP532 – Bioinspired Learning
University of Liverpool

Problem Definition

- Binary Classification task
 - Labels: 0 and 1
- Multilayer Perceptron architecture:
 - One input layer
 - Two input nodes
 - Two hidden layers
 - Activation function: sigmoid
 - One output layer
 - Activation function: sigmoid
- Aim: update a connection weight using backprop
 - Loss function: binary cross-entropy (BCE)
- No bias terms in this example

Problem Definition

- Input pattern: 1 and -2
- Learning rate:

$$\mathcal{C} = 0.1$$

- Output label:

$$y = 1$$

- Output neurons p_1 and p_2 represent labels 1 and 0, respectively
- Weight to update: $w_{1.5}$ (see diagram)

Training Process (Recap)

1. Calculate the weighted sum of each neuron in the first hidden layer
2. Pass the weighted sum through an activation function
 - Repeat steps 1 and 2 until the output layer is reached
3. Calculate the error
 1. Calculate the error using an appropriate function for the task
 2. Find the gradient of the error function
 3. Calculate the error signal
4. Backpropagate the error to update weights

Formulae

- Weighted sum (x) and post-activation value ($\hat{y}$)

$$x = \sum_{i=1}^n w_i x_i + b$$

$$\hat{y} = f(x)$$

- Activation function (sigmoid)

$$f(x) = \frac{1}{1+e^{-x}}$$

- Error E (BCE)

$$E = -[y \log \hat{y} + (1 - y) \log(1 - \hat{y})]$$

- Weight update rule using stochastic gradient descent (SGD)

$$w_i^{t+1} = w_i^t - C \frac{\partial E}{\partial w_i^t}$$

Derivatives

- Activation function

$$f(x) = \frac{1}{1 + e^{-x}}$$

$$f'(x) = f(x)(1 - f(x))$$

$$\frac{\partial \hat{y}}{\partial x} = \hat{y}(1 - \hat{y})$$

- Loss function

$$\frac{\partial E}{\partial \hat{y}} = \frac{\hat{y} - y}{\hat{y}(1 - \hat{y})} \cdot \hat{y}(1 - \hat{y}) = \hat{y} - y$$

- Derivative of the loss E with respect to weight w_i

$$\frac{\partial E}{\partial w_i} = (\hat{y} - y) \cdot \hat{y}(1 - \hat{y}) \cdot x_i$$

- Note: x_i is the input value from the input node/neuron

Chain Rule

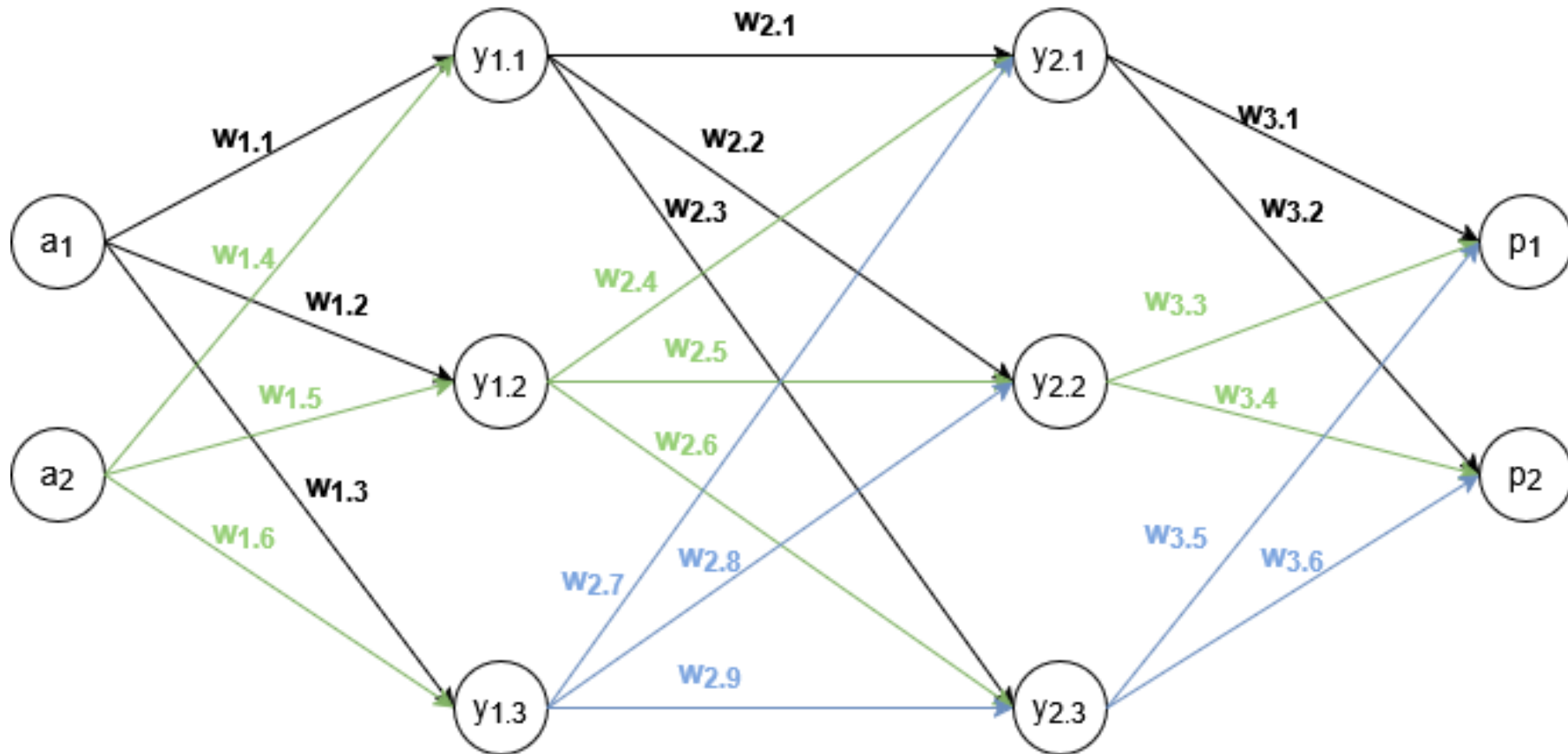
- We can update a weight using the Chain Rule
- To update $w_{1.5}$, we need to use the following chain:

$$\frac{\partial E}{\partial w_{1.5}} = \sum_j \frac{\partial E}{\partial p_j} \cdot \frac{\partial p_j}{\partial y_{2.k}} \cdot \frac{\partial y_{2.k}}{\partial y_{1.2}} \cdot \frac{\partial y_{1.2}}{\partial w_{1.5}}$$

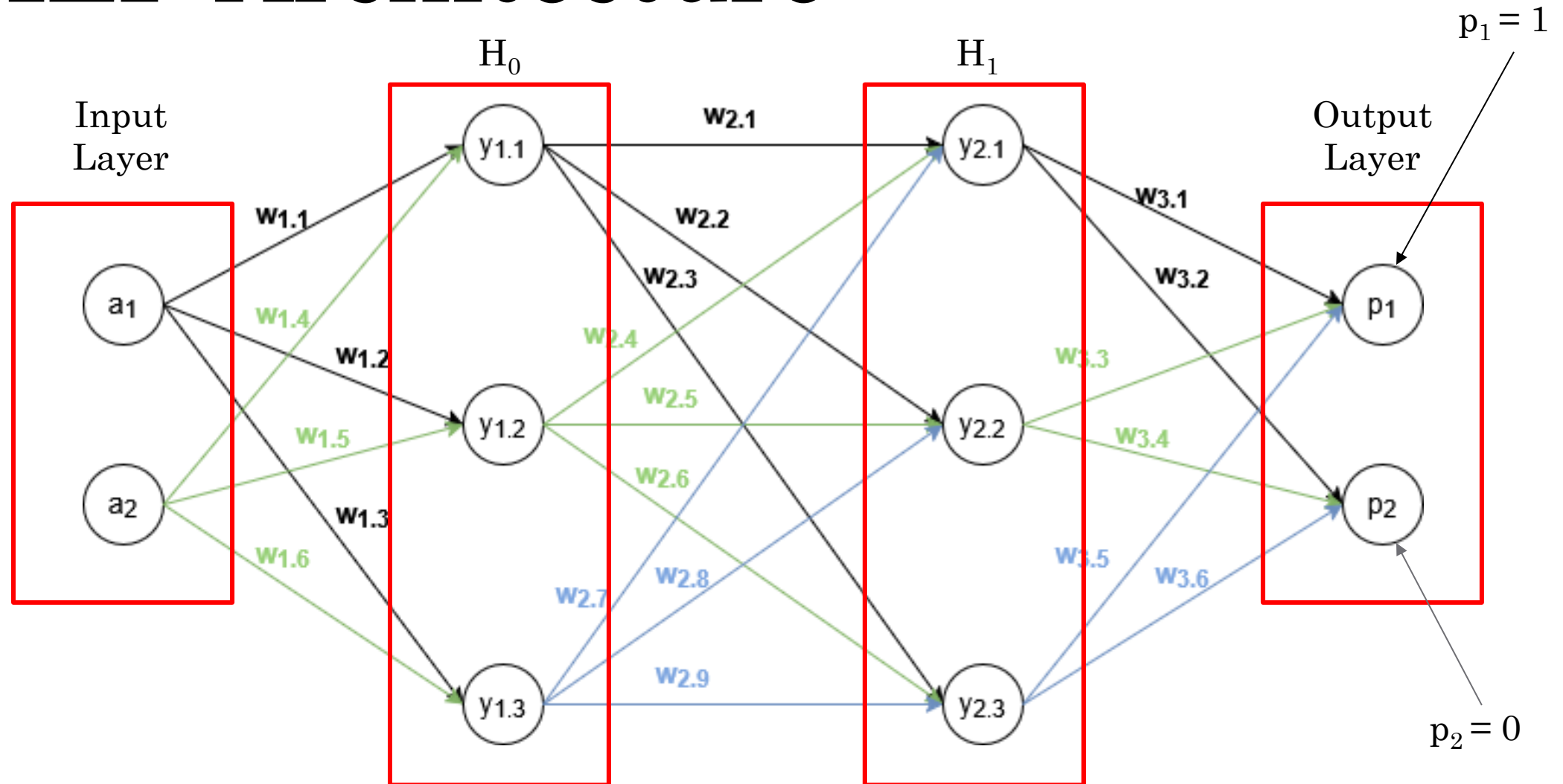
Where:

- $\frac{\partial E}{\partial p_j}$ is the derivative of the error of the output layer
- $\frac{\partial p_j}{\partial y_{2.k}}$ is the derivative of the activation function of neuron k in the second hidden layer
- $\frac{\partial y_{2.k}}{\partial y_{1.2}}$ is the derivative of the activation function of neuron 2 in the first hidden layer
- $\frac{\partial y_{1.2}}{\partial w_{1.5}}$ is the derivative of the weighted sum of neuron 2 in the first hidden layer w.r.t weight $w_{1.5}$ (we reach the input layer at this stage)

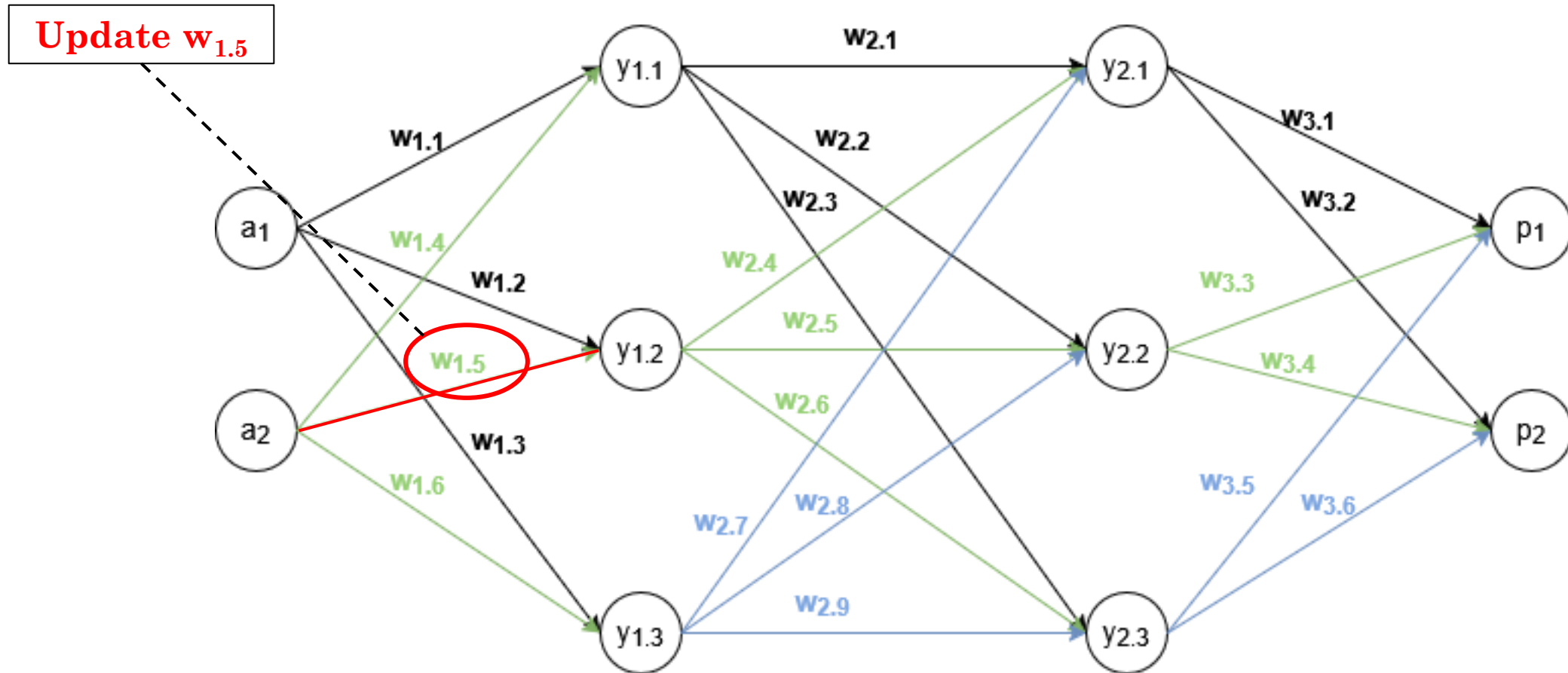
MLP Architecture



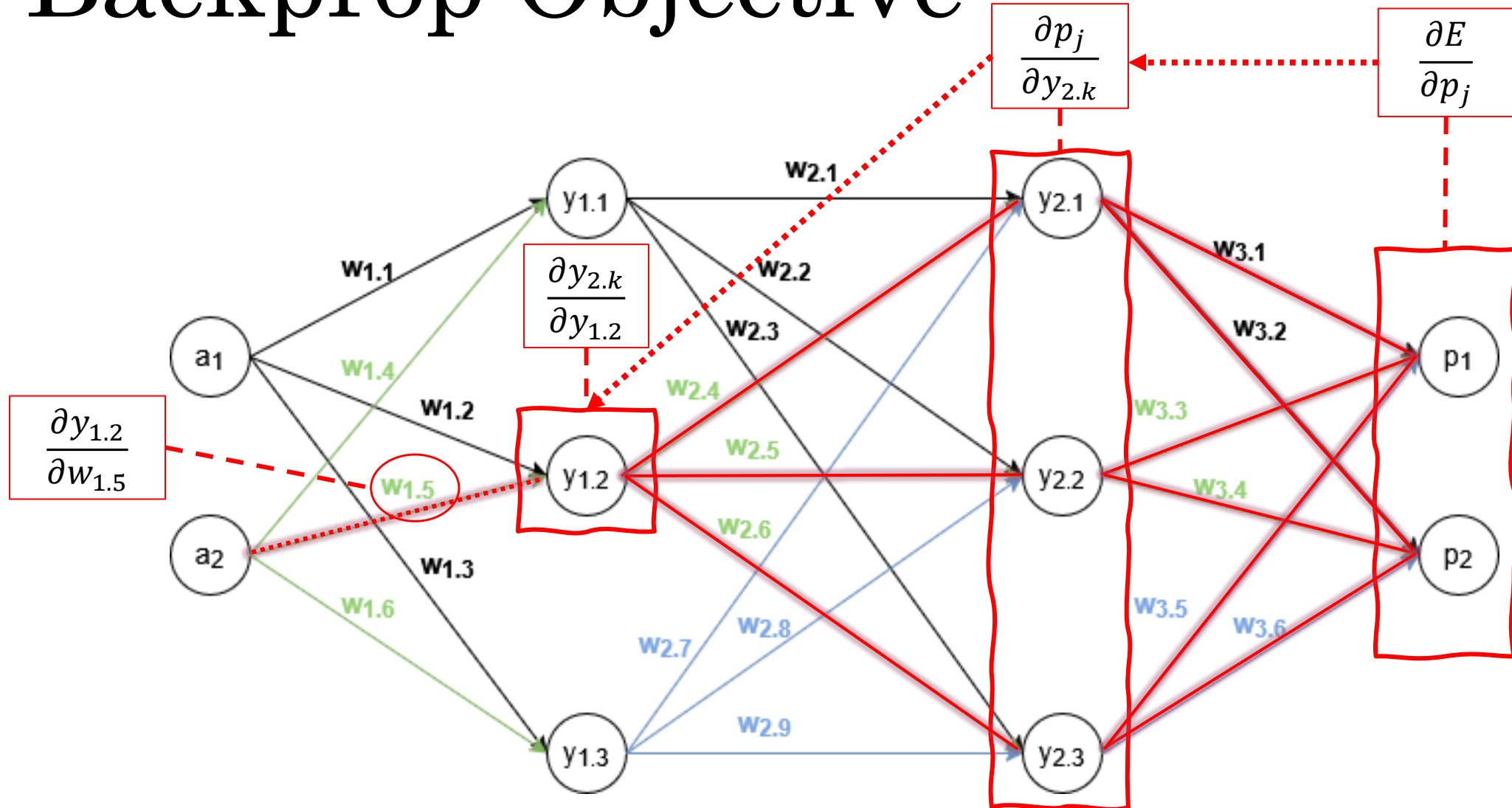
MLP Architecture



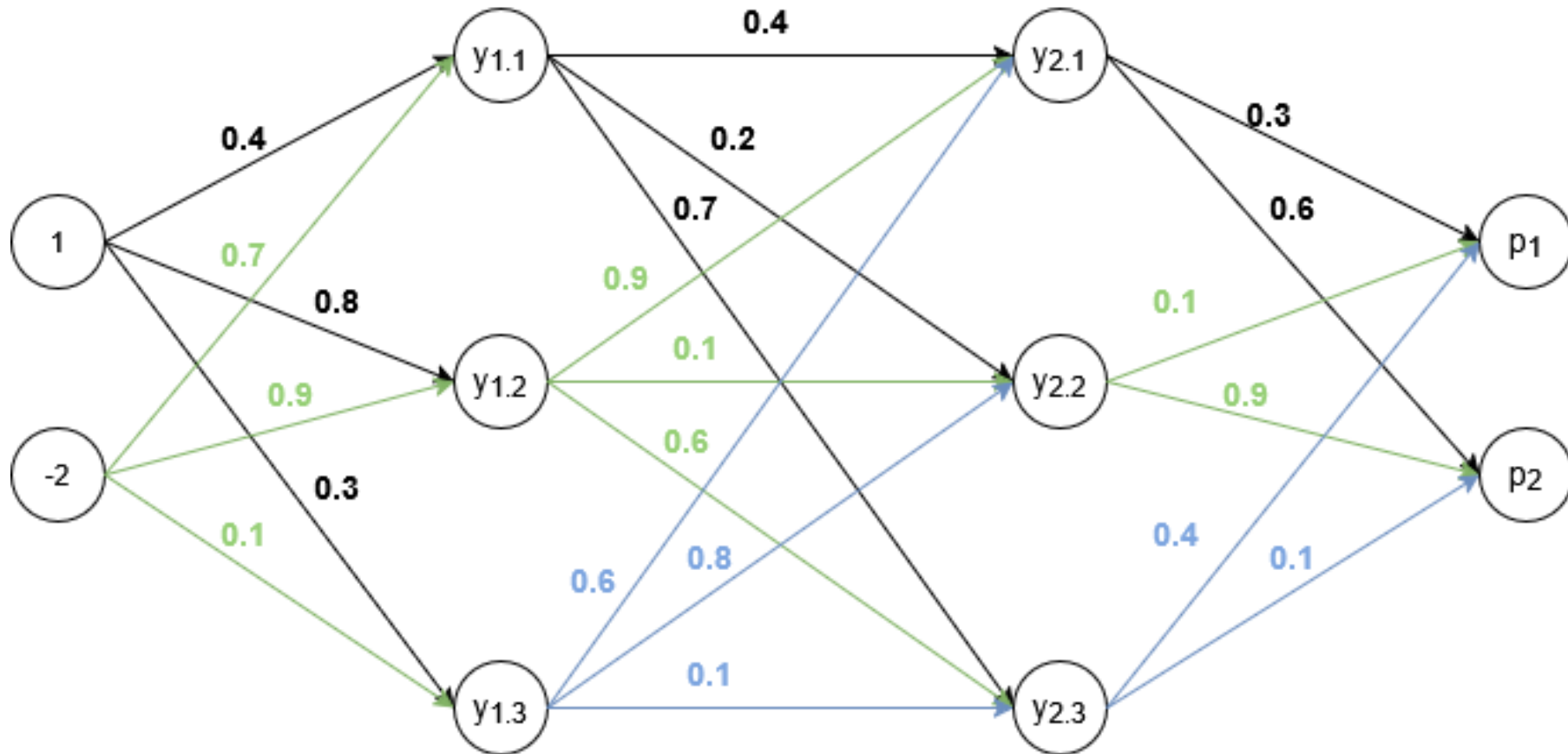
Backprop Objective



Backprop Objective



MLP (Randomised Weights)



Step 1: Weighted Sums (H_0)

- $y_{1.1} = a_1 w_{1.1} + a_2 w_{1.4}$

- $y_{1.2} = a_1 w_{1.2} + a_2 w_{1.5}$

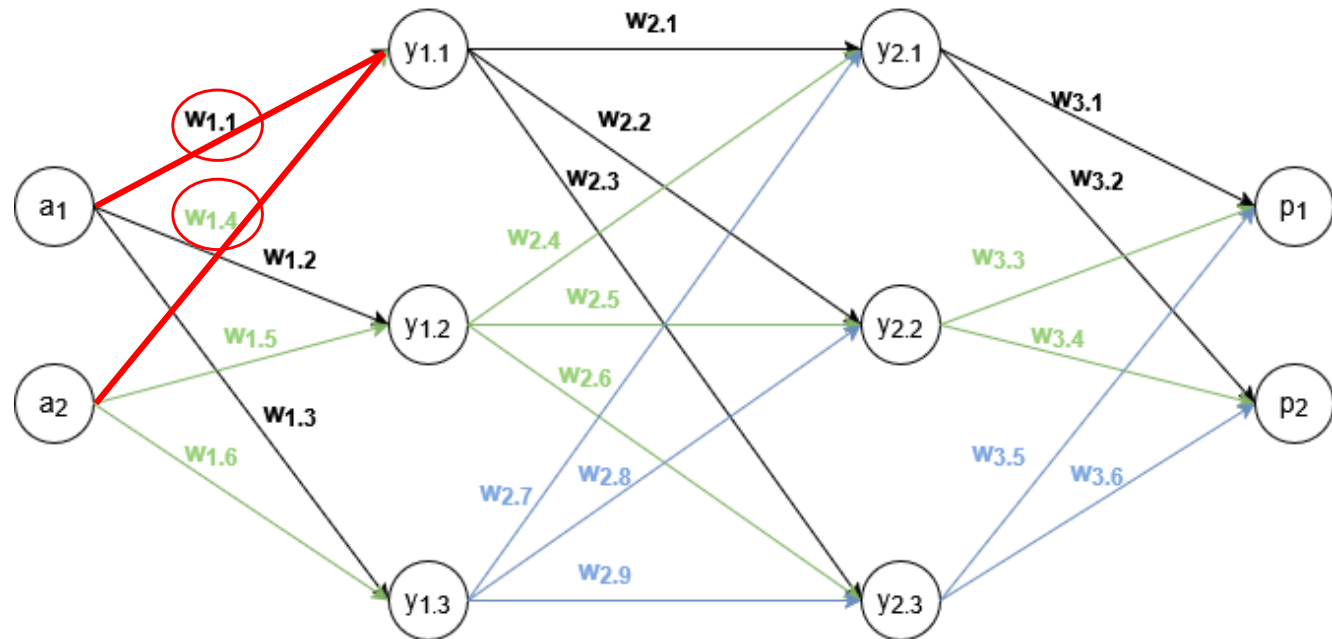
- $y_{1.3} = a_1 w_{1.3} + a_2 w_{1.6}$

- $y_{1.1} = 1 \times 0.4 + (-2) \times 0.7 = -1$

$$a_1 = 1$$
$$a_2 = -2$$

$$w_{1.1} = 0.4$$
$$w_{1.4} = 0.7$$

$$y_n = x = \sum_{i=1}^n w_i x_i + b$$



Step 1: Weighted Sums (H_0)

- $y_{1.1} = a_1 w_{1.1} + a_2 w_{1.4}$

- $y_{1.2} = a_1 w_{1.2} + a_2 w_{1.5}$

- $y_{1.3} = a_1 w_{1.3} + a_2 w_{1.6}$

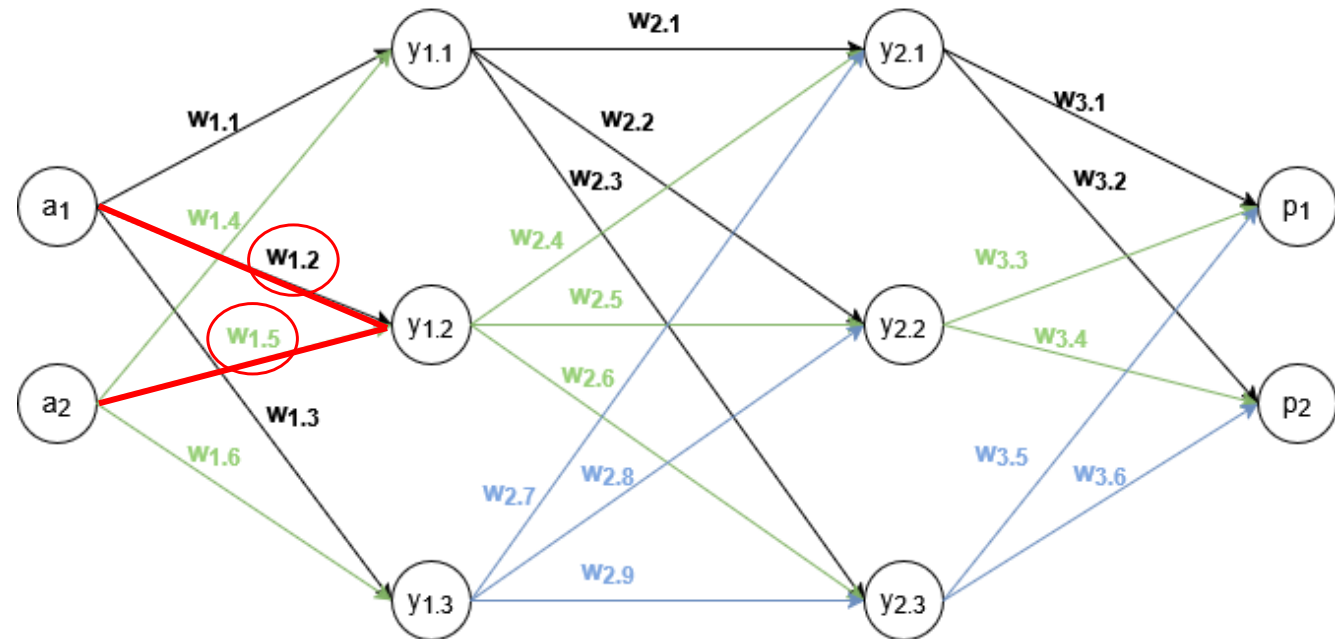
$$a_1 = 1$$
$$a_2 = -2$$

$$w_{1.2} = 0.8$$
$$w_{1.5} = 0.9$$

$$y_n = x = \sum_{i=1}^n w_i x_i + b$$

- $y_{1.1} = 1 \times 0.4 + (-2) \times 0.7 = -1$

- $y_{1.2} = 1 \times 0.8 + (-2) \times 0.9 = -1$



Step 1: Weighted Sums (H_0)

- $y_{1.1} = a_1 w_{1.1} + a_2 w_{1.4}$

- $y_{1.2} = a_1 w_{1.2} + a_2 w_{1.5}$

- $y_{1.3} = a_1 w_{1.3} + a_2 w_{1.6}$

$$a_1 = 1$$
$$a_2 = -2$$

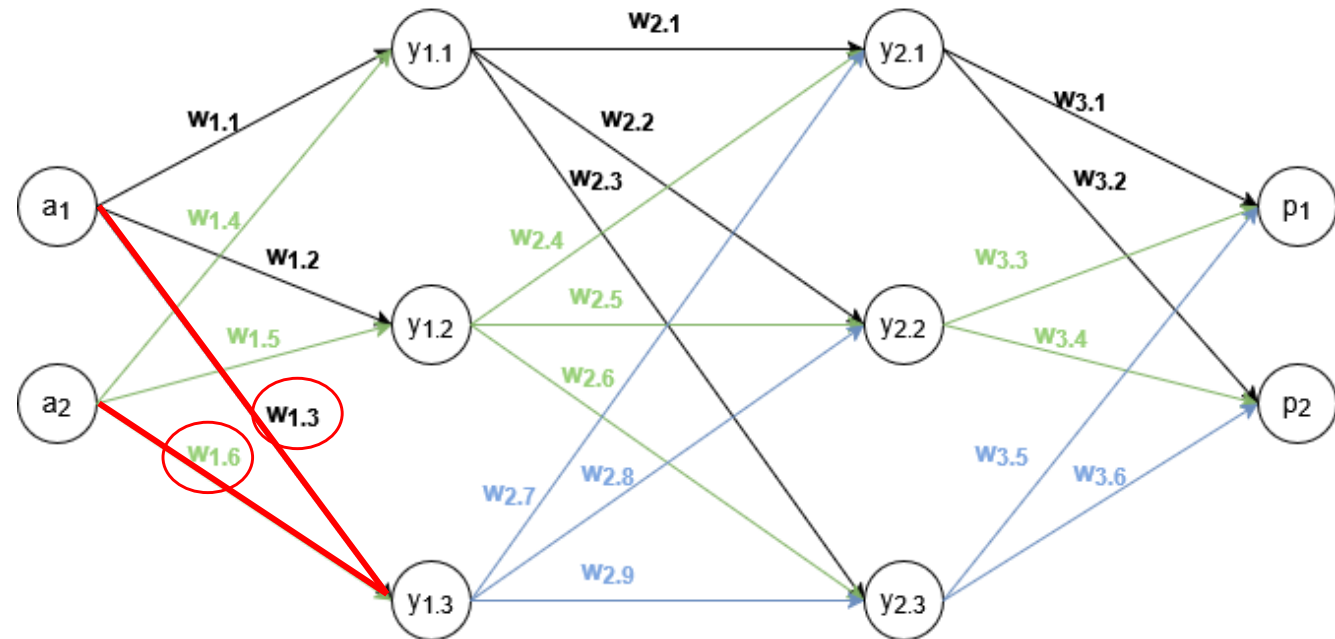
$$w_{1.3} = 0.3$$
$$w_{1.6} = 0.1$$

$$y_n = x = \sum_{i=1}^n w_i x_i + b$$

- $y_{1.1} = 1 \times 0.4 + (-2) \times 0.7 = -1$

- $y_{1.2} = 1 \times 0.8 + (-2) \times 0.9 = -1$

- $y_{1.3} = 1 \times 0.3 + (-2) \times 0.1 = 0.1$



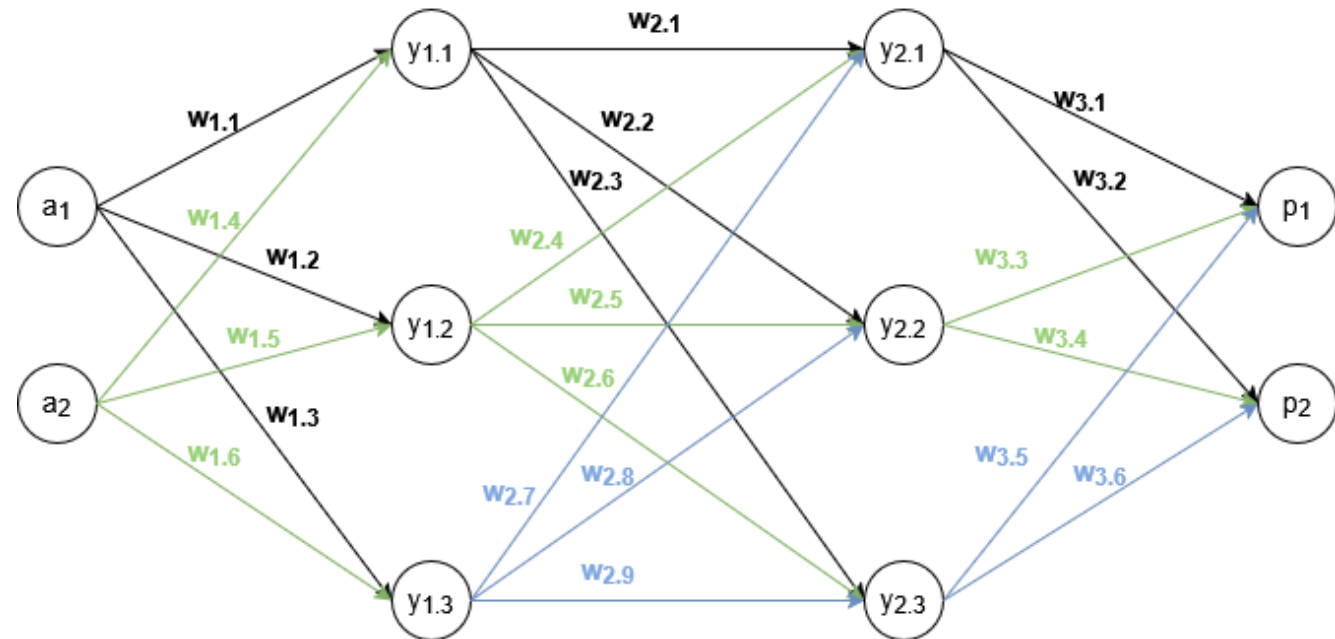
Step 2: Activation (H_0)

- $y_{1.1} = a_1 w_{1.1} + a_2 w_{1.4}$
- $y_{1.2} = a_1 w_{1.2} + a_2 w_{1.5}$
- $y_{1.3} = a_1 w_{1.3} + a_2 w_{1.6}$

$$\begin{aligned}y_{1.1} &= -1 \\ y_{1.2} &= -1 \\ y_{1.3} &= 0.1\end{aligned}$$

$$\hat{y} = f(x) = \frac{1}{1 + e^{-x}}$$

- $f(y_{1.1}) = \frac{1}{1 + e^{-(-1)}} = 0.2689$
- $f(y_{1.2}) = \frac{1}{1 + e^{-(-1)}} = 0.2689$
- $f(y_{1.3}) = \frac{1}{1 + e^{-0.1}} = 0.5250$



Step 2: Activation (H_0)

- $f(y_{1.1}) = \frac{1}{1+e^{-y_{1.1}}}$

- $f(y_{1.2}) = \frac{1}{1+e^{-y_{1.2}}}$

- $f(y_{1.3}) = \frac{1}{1+e^{-y_{1.3}}}$

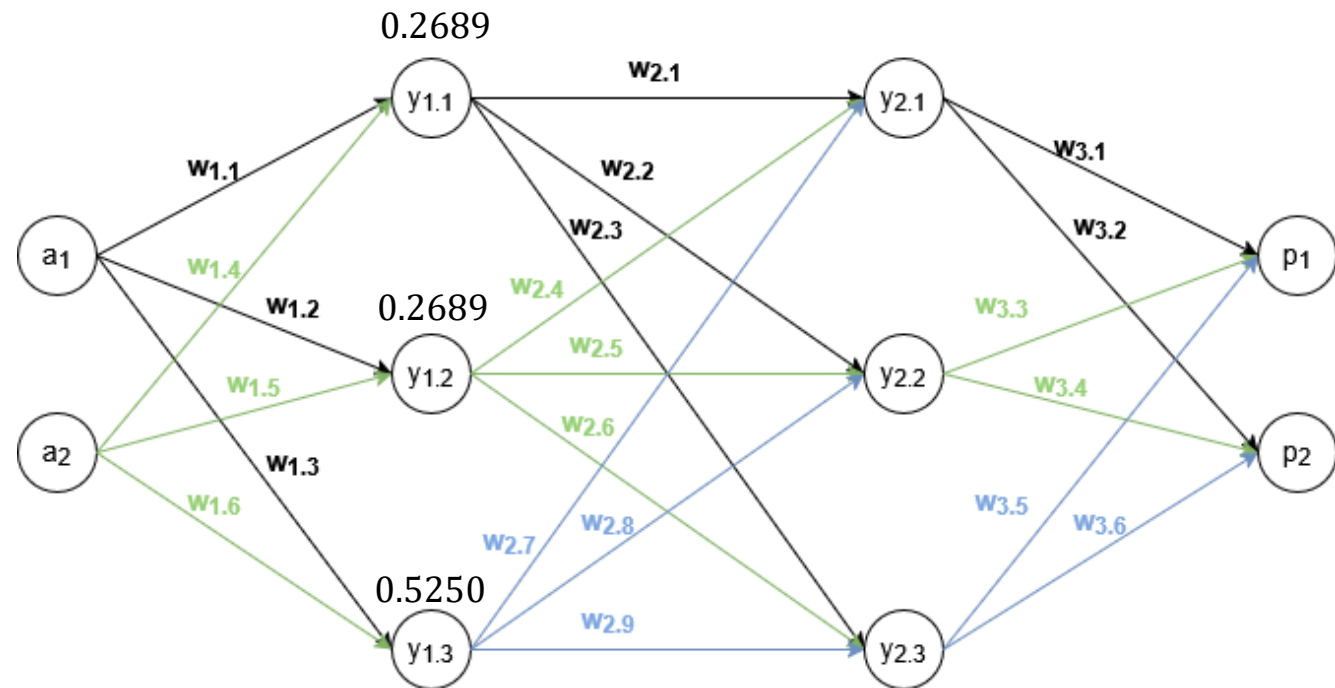
$$\begin{aligned} y_{1.1} &= -1 \\ y_{1.2} &= -1 \\ y_{1.3} &= 0.1 \end{aligned}$$

$$f(x) = \frac{1}{1 + e^{-x}}$$

- $f(y_{1.1}) = \frac{1}{1+e^{-(-1)}} = 0.2689$

- $f(y_{1.2}) = \frac{1}{1+e^{-(-1)}} = 0.2689$

- $f(y_{1.2}) = \frac{1}{1+e^{-0.1}} = 0.5250$

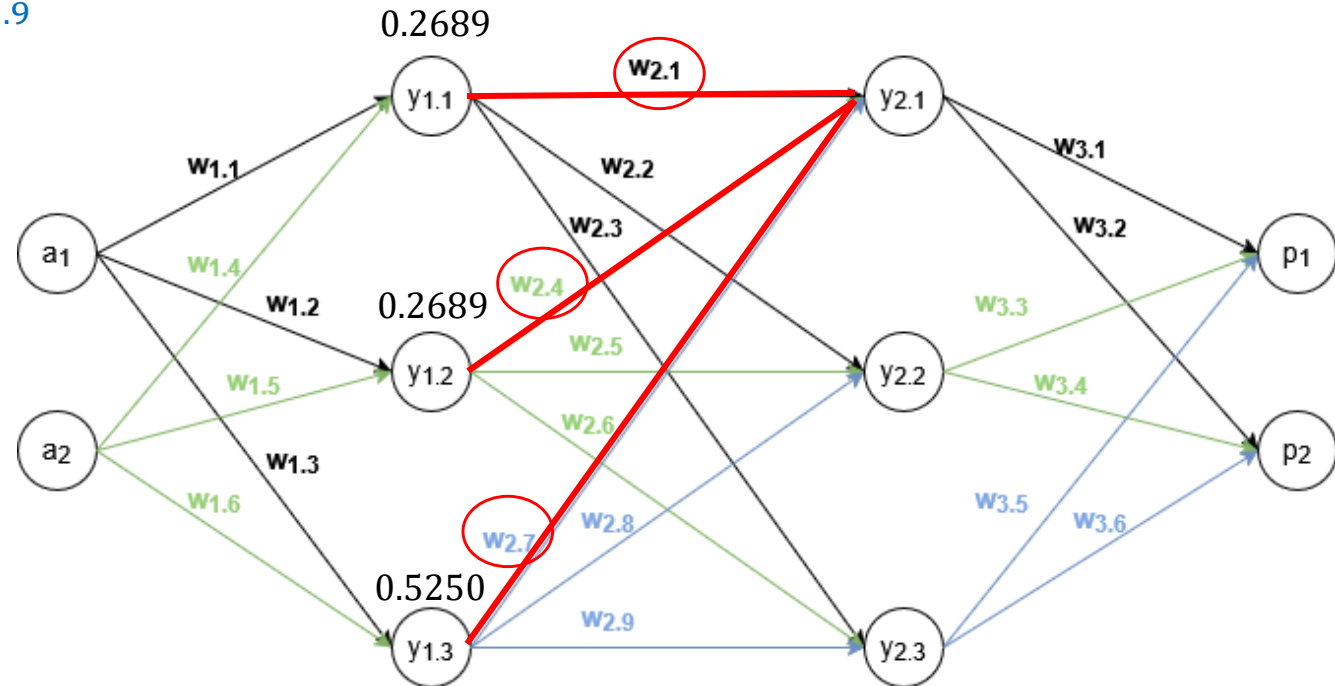


Repeat for H_1

Step 1: Weighted Sums (H_1)

- $y_{2.1} = y_{1.1}w_{2.1} + y_{1.2}w_{2.4} + y_{1.3}w_{2.7}$
 - $y_{2.2} = y_{1.1}w_{2.2} + y_{1.2}w_{2.5} + y_{1.3}w_{2.8}$
 - $y_{2.3} = y_{1.1}w_{2.3} + y_{1.2}w_{2.6} + y_{1.3}w_{2.9}$
- $y_{1.1} = 0.2689$ $w_{2.1} = 0.4$
 $y_{1.2} = 0.2689$ $w_{2.4} = 0.9$
 $y_{1.3} = 0.5250$ $w_{2.7} = 0.6$

$$y_n = x = \sum_{i=1}^n w_i x_i + b$$



Step 1: Weighted Sums (H_1)

- $y_{2.1} = y_{1.1}w_{2.1} + y_{1.2}w_{2.4} + y_{1.3}w_{2.7}$

$$y_{1.1} = 0.2689$$

$$w_{2.1} = 0.4$$

$$y_{1.2} = 0.2689$$

$$w_{2.4} = 0.9$$

$$y_{1.3} = 0.5250$$

$$w_{2.7} = 0.6$$

- $y_{2.3} = y_{1.1}w_{2.3} + y_{1.2}w_{2.6} + y_{1.3}w_{2.9}$

$$\bullet y_{2.1} = 0.2689 \times 0.4 + 0.2689 \times 0.9 + 0.5250 \times 0.9 = 0.6646$$

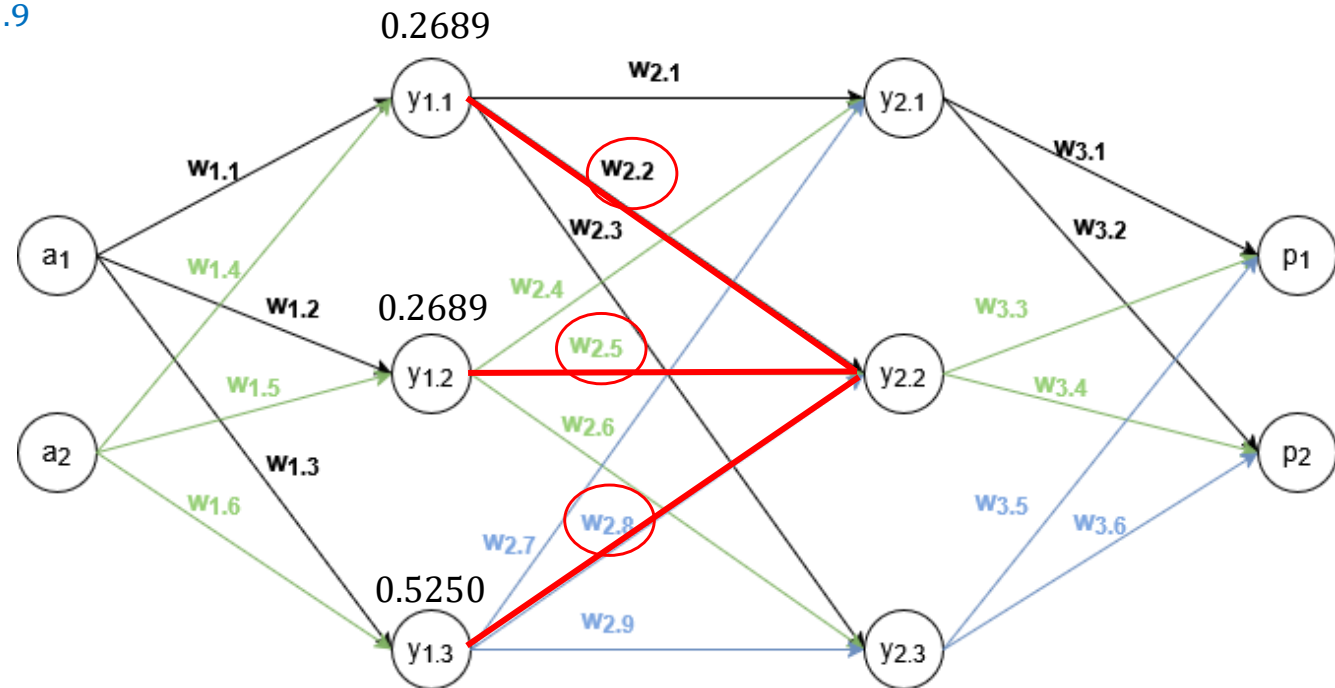
$$y_n = x = \sum_{i=1}^n w_i x_i + b$$

Step 1: Weighted Sums (H_1)

- $y_{2.1} = y_{1.1}w_{2.1} + y_{1.2}w_{2.4} + y_{1.3}w_{2.7}$ $y_{1.1} = 0.2689$ $w_{2.2} = 0.2$
- $y_{2.2} = y_{1.1}w_{2.2} + y_{1.2}w_{2.5} + y_{1.3}w_{2.8}$ $y_{1.2} = 0.2689$ $w_{2.5} = 0.1$
 $y_{1.3} = 0.5250$ $w_{2.8} = 0.8$
- $y_{2.3} = y_{1.1}w_{2.3} + y_{1.2}w_{2.6} + y_{1.3}w_{2.9}$

$$y_n = x = \sum_{i=1}^n w_i x_i + b$$

- $y_{2.1} = 0.6646$
- $y_{2.2} = 0.5007$

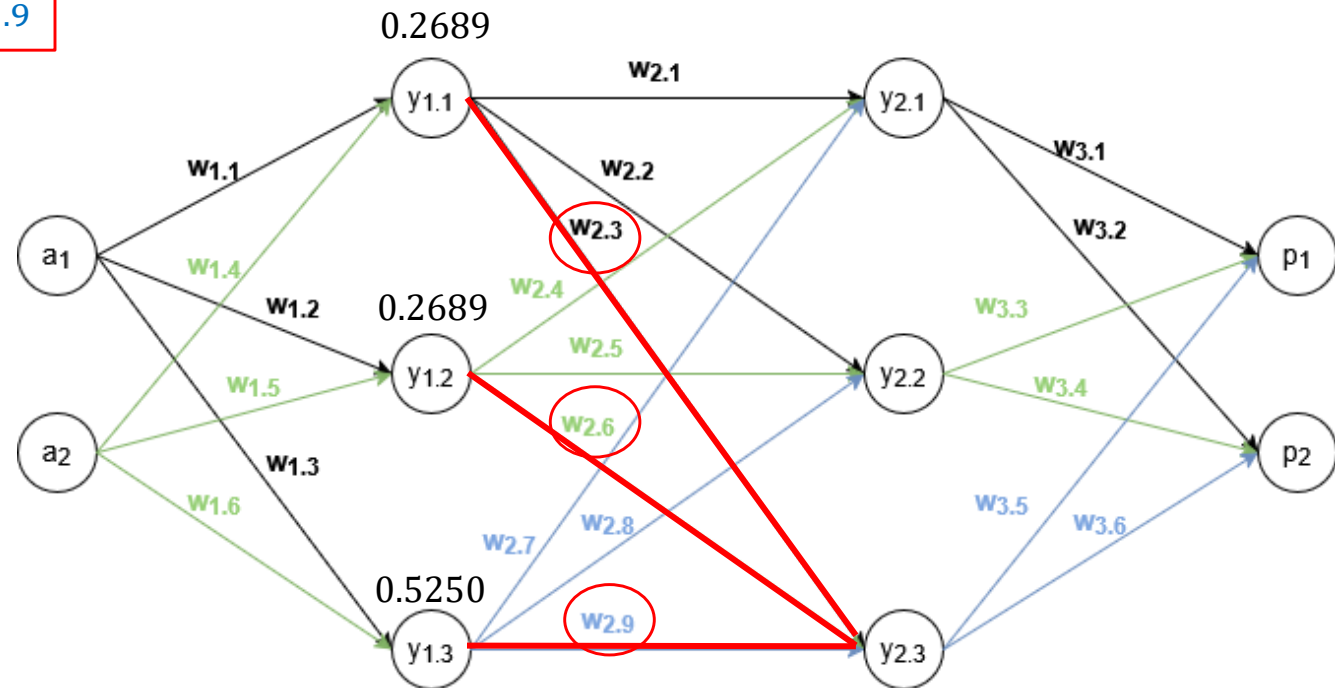


Step 1: Weighted Sums (H_1)

- $y_{2.1} = y_{1.1}w_{2.1} + y_{1.2}w_{2.4} + y_{1.3}w_{2.7}$ $y_{1.1} = 0.2689$ $w_{2.3} = 0.7$
- $y_{2.2} = y_{1.1}w_{2.2} + y_{1.2}w_{2.5} + y_{1.3}w_{2.8}$ $y_{1.2} = 0.2689$ $w_{2.6} = 0.6$
- $y_{2.3} = y_{1.1}w_{2.3} + y_{1.2}w_{2.6} + y_{1.3}w_{2.9}$ $y_{1.3} = 0.5250$ $w_{2.9} = 0.1$

$$y_n = x = \sum_{i=1}^n w_i x_i + b$$

- $y_{2.1} = 0.6646$
- $y_{2.2} = 0.5007$
- $y_{2.3} = 0.4021$



Step 2: Activation (H_1)

- $f(y_{1.1}) = \frac{1}{1+e^{-y_{2.1}}}$

- $f(y_{2.2}) = \frac{1}{1+e^{-y_{2.2}}}$

- $f(y_{2.3}) = \frac{1}{1+e^{-y_{2.3}}}$

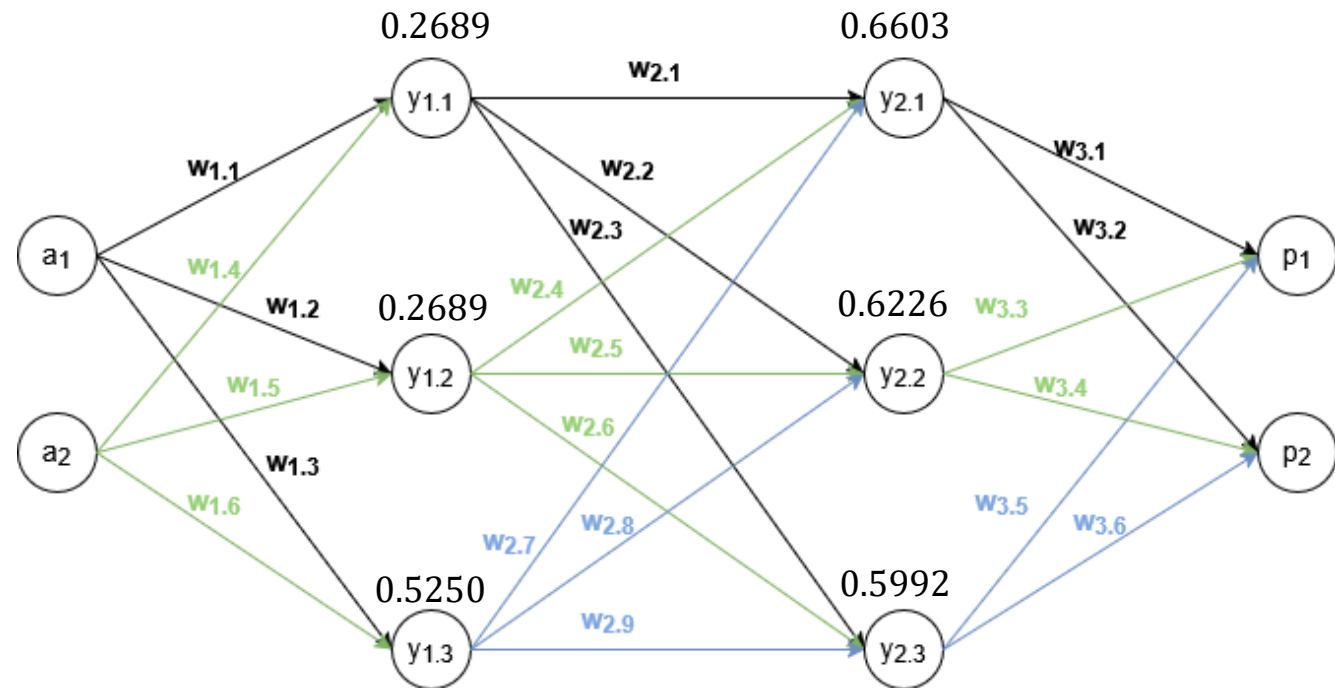
$$\begin{aligned} y_{2.1} &= 0.6646 \\ y_{2.2} &= 0.5007 \\ y_{2.3} &= 0.4021 \end{aligned}$$

$$f(x) = \frac{1}{1 + e^{-x}}$$

- $f(y_{2.1}) = \frac{1}{1+e^{-0.6646}} = 0.6603$

- $f(y_{2.2}) = \frac{1}{1+e^{-0.5007}} = 0.6226$

- $f(y_{2.2}) = \frac{1}{1+e^{-0.4021}} = 0.5992$

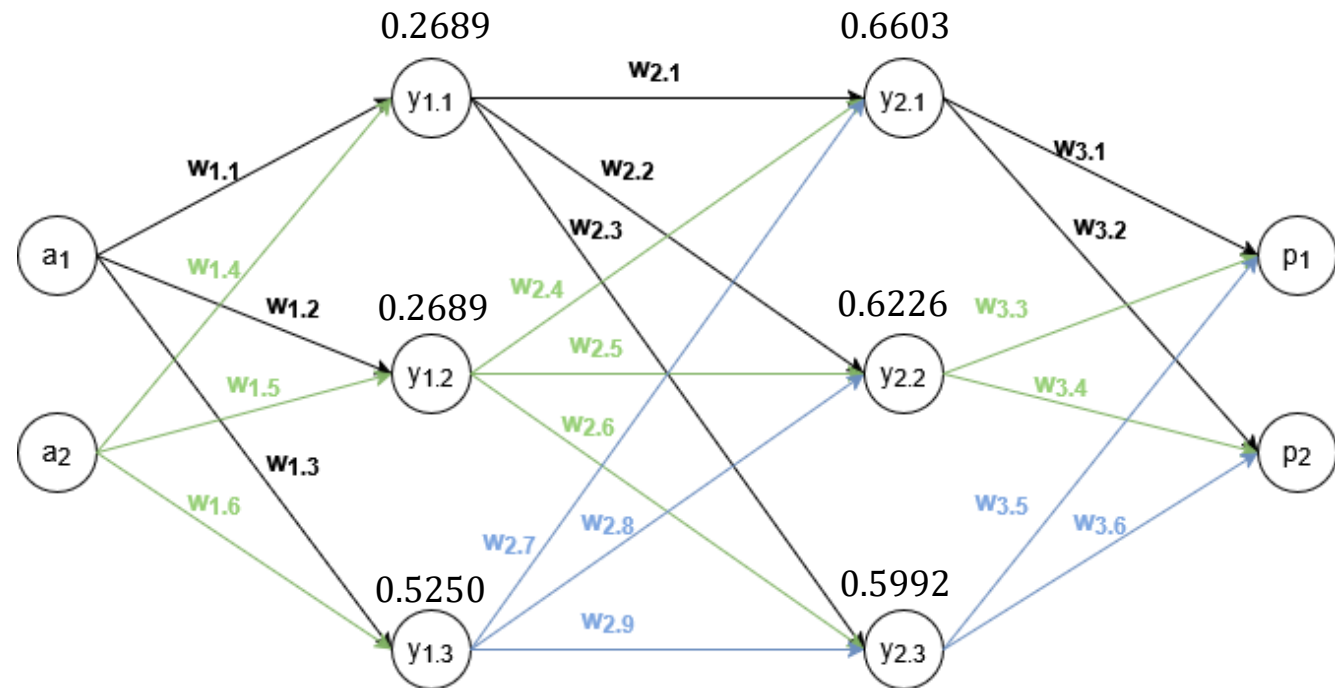


Repeat for H_2 (Output Layer)

Step 1& 2 (Output Layer)

- $p_1 = 0.4704$
- $p_2 = 1.0090$
- $f(p_1) = 0.6155$
- $f(p_2) = 0.7328$

$$x = \sum_{i=1}^n w_i x_i + b$$
$$\hat{y} = f(x) = \frac{1}{1 + e^{-x}}$$



Step 3.1: Error Calculation

- $p_1 = 0.4704$
- $p_2 = 1.0090$
- $f(p_1) = 0.6155$
- $f(p_2) = 0.7328$
- For now, we ignore the BCE formula. Instead, we're interested in its derivative, $\frac{\partial E}{\partial \hat{y}}$.
 - It is still useful for monitoring convergence. For training purposes, we are interested in the derivative.
- We need to calculate the derivative of each output neuron's error. Generally, the formula is the following:

$$\frac{\partial E_{p_n}}{\partial p_n} = p_n - y_n$$

Where:

- p_n is the prediction produced by the output neuron n
- y_n is the label output neuron n represents (in our example, either 1 or 0)

$$E = -[y \log \hat{y} + (1 - y) \log(1 - \hat{y})]$$

$$\frac{\partial E}{\partial \hat{y}} = \hat{y} - y$$

Step 3.1: Error Calculation

$$\frac{\partial E}{\partial \hat{y}} = \hat{y} - y$$

- $f(p_1) = 0.6155$

- $f(p_2) = 0.7328$

- Derivative of the error of p_1 (representing 1)

$$\frac{\partial E_{p_1}}{\partial p_1} = p_1 - y_1 = 0.6155 - 1 = -0.3845$$

- Derivative of the error of p_2 (representing 0)

$$\frac{\partial E_{p_2}}{\partial p_2} = p_2 - y_2 = 0.7328 - 0 = 0.7328$$

Step 3.2: Finding the Gradients

$$\frac{\partial \hat{y}}{\partial x} = \hat{y}(1 - \hat{y})$$

- $f(p_1) = 0.6155, \frac{\partial E_{p_1}}{\partial p_1} = -0.3845$

- $f(p_2) = 0.7328, \frac{\partial E_{p_2}}{\partial p_2} = 0.7328$

- Gradient of $f(p_1)$:

$$\frac{\partial p_1}{\partial x} = p_1(1 - p_1) = 0.6155(1 - 0.6155) = 0.2366$$

- Gradient of $f(p_2)$:

$$\frac{\partial p_2}{\partial x} = p_2(1 - p_2) = 0.7328(1 - 0.7328) = 0.1956$$

Step 3.3: Getting the Error Signal

$$\partial p_n = \frac{\partial E_{p_n}}{\partial p_n} \cdot \frac{\partial p_n}{\partial x}$$

- $f(p_1) = 0.6155, \frac{\partial E_{p_1}}{\partial p_1} = -0.3845, \frac{\partial p_1}{\partial x} = 0.2366$

- $f(p_2) = 0.7328, \frac{\partial E_{p_2}}{\partial p_2} = 0.7328, \frac{\partial p_2}{\partial x} = 0.1956$

- Error Signal of $f(p_1)$:

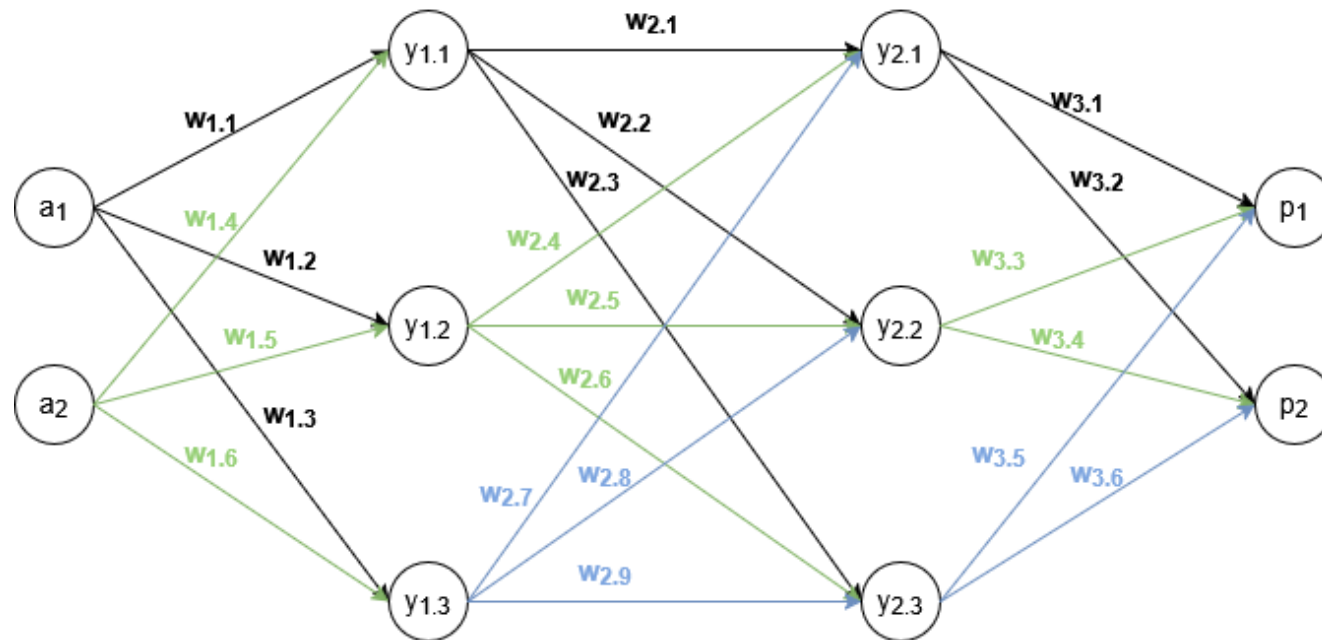
$$\frac{\partial E}{\partial p_1} = \frac{\partial E_{p_1}}{\partial p_1} \cdot \frac{\partial p_1}{\partial x} = (p_1 - y_1) \cdot p_1(1 - p_1) = -0.3845 \times 0.2366 = -0.0910$$

- Error Signal of $f(p_2)$:

$$\frac{\partial E}{\partial p_2} = \frac{\partial E_{p_2}}{\partial p_2} \cdot \frac{\partial p_2}{\partial x} = (p_2 - y_2) \cdot p_2(1 - p_2) = 0.7328 \times 0.1956 = 0.1434$$

Step 4: Backprop

- We want to update $w_{1.5}$ in this exercise
- To do this, we need to backpropagate the error through the layers
 - During this process, the weight of each connection we traverse is updated using the chain rule
- Then, we update the weight using SGD



Step 4.1: Chain Rule

- Specific to our example, the chain rule can be defined as:

$$\frac{\partial E}{\partial w_{1.5}} = \sum_j \frac{\partial E}{\partial p_j} \cdot \frac{\partial p_j}{\partial y_{2.k}} \cdot \frac{\partial y_{2.k}}{\partial y_{1.2}} \cdot \frac{\partial y_{1.2}}{\partial w_{1.5}}$$

- We can also express this as:

$$\frac{\partial E}{\partial w_{1.5}} = \sum_j (p_j - y_j) \cdot p_j(1 - y_j) \cdot w_{3.k} \cdot y_{2.k}(1 - y_{2.k}) \cdot a_2$$

Step 4.1: Chain Rule

- From 3.1, we have:

$$\frac{\partial E}{\partial p_1} = -0.3845 \text{ and } \frac{\partial E}{\partial p_2} = 0.7328$$

- From 3.2, we have:

$$\frac{\partial p_1}{\partial x} = 0.2336 \text{ and } \frac{\partial p_2}{\partial x} = 0.1956$$

- From 3.3, we have:

$$\frac{\partial E}{\partial p_1} = -0.0910 \text{ and } \frac{\partial E}{\partial p_2} = 0.1434$$

- We need to propagate from layer 2 to layer 1
 - More specifically, we need to propagate the error to neuron $y_{1,2}$

Step 4.2: Between Hidden Layers

- $y_{2.1} = 0.6646, y_{2.2} = 0.5007, y_{2.3} = 0.4021$
- $\partial p_1 = -0.0910, \partial p_2 = 0.1434$
- We need to calculate the error signal of each hidden neuron (just like we did in 3.3)

$$\frac{\partial E}{\partial y_{2.k}} = \sum_j \left(\frac{\partial E}{\partial p_j} \cdot w_{3.k} \right) \cdot f'(y_{2.k})$$

- $\frac{\partial E}{\partial y_{2.1}} = (-0.0910 \times 0.3 + 0.1434 \times 0.6) \times 0.6646(1 - 0.6646) = 0.0131$
- $\frac{\partial E}{\partial y_{2.2}} = (-0.0910 \times 0.1 + 0.1434 \times 0.9) \times 0.5007(1 - 0.5007) = 0.0010$
- $\frac{\partial E}{\partial y_{2.3}} = (-0.0910 \times 0.4 + 0.1434 \times 0.1) \times 0.4021(1 - 0.4021) = -0.005$

Step 4.2: Between Hidden Layers

- We have the error signals for hidden layer 2, now we can propagate to layer 1. We need to do this for the neuron $y_{1.2}$ as this is the neuron that's connected to the input. We will use the same formula as before.
- $w_{2.4} = 0.9, w_{2.5} = 0.1, w_{2.6} = 0.6$
- $\frac{\partial E}{\partial y_{2.1}} = 0.0131, \frac{\partial E}{\partial y_{2.2}} = 0.0010, \frac{\partial E}{\partial y_{2.3}} = -0.005,$
- $\frac{\partial E}{\partial y_{1.2}} = (0.0131 \times 0.9 + 0.0010 \times 0.1 + (-0.005 \times 0.6)) \times 0.2689(1 - 0.2689) = 0.0017$

Step 4.2: Reaching the Input

- We can now calculate the final term of the chain

$$\frac{\partial E}{\partial w_{1.5}} = \frac{\partial E}{\partial y_{1.2}} \cdot a_2$$

- $\frac{\partial E}{\partial w_{1.5}} = 0.0017 \cdot (-2) = -0.0035$

Step 5: SGD

- To update the weight, we can use stochastic gradient descent

$$w_i^{t+1} = w_i^t - C \frac{\partial E}{\partial w_i^t}$$

- From 4.2, we have:

- $\frac{\partial E}{\partial w_{1.5}} = -0.0035$

- Reminder: $C = 0.1$ and $w_{1.5} = 0.9$

- $w_{1.5}^{t+1} = 0.9 - 0.1 \times (-0.0035) = 0.9004$